10.11 (a) The incident wave can be expressed as $\psi_i(\vec{x}) = I_0^{1/2} e^{ikz - i\omega t}$. Applying Eq. (10.85), the diffracted field is

$$\psi(\vec{x}, t) = \frac{k}{2\pi i} \int_{S_1} \frac{e^{ikR}}{R}\left(1 + \frac{i}{kR}\right)\frac{\vec{n}'\cdot\vec{R}}{R}\,\psi(\vec{x}')\,da',$$

where $\vec{R} = (X - x', -y', Z)$, and the above expression becomes

$$\psi(X, 0, Z, t) = \frac{k}{2\pi i}\int_{-\infty}^{+\infty} dy' \int_0^{+\infty} dx' \frac{\exp\{ik\sqrt{(X-x')^2 + y'^2 + Z^2}\}}{\sqrt{(X-x')^2 + y'^2 + Z^2}}\left(1 + \frac{i}{k\sqrt{(X-x')^2 + y'^2 + Z^2}}\right)$$

$$\times \frac{Z}{\sqrt{(X-x')^2 + y'^2 + Z^2}}\, e^{-i\omega t} I_0^{1/2}$$

We can use the following integration identity

$$\int_0^{+\infty}\left(\frac{1}{\alpha(x^2+b^2)^{3/2}} + \frac{1}{x^2+b^2}\right)\exp\{-\alpha\sqrt{x^2+b^2}\}\cos(\gamma x)\,dx = \frac{\sqrt{\alpha^2+\gamma^2}}{\alpha b} K_1\left(b\sqrt{\alpha^2+\gamma^2}\right).$$

to perform the integration in the y'-axis by setting $\alpha = -ik$, $\gamma = 0$, $b = \sqrt{(X-x')^2 + Z^2}$ to obtain

$$\psi(X, 0, Z, t) = \frac{k}{\pi i}\int_0^{+\infty} dx' \frac{Z K_1\left(-ik\sqrt{(X-x')^2 + Z^2}\right)}{\sqrt{(X-x')^2 + Z^2}}\, e^{-i\omega t} I_0^{1/2}$$

Here, we are quite sloppy in hadling square root of negative number. I don't how to further process the integral rigorously and therefore have to resort to approximations. For large Z with $kZ \gg 1$, we can use $K_\nu(x) \to \sqrt{\frac{\pi}{2x}}\, e^{-x}$, $x \to \infty$, to write the integral as

$$\psi(x, 0, Z, t) = \frac{k}{\pi i}\int_0^{+\infty} dx' \sqrt{\frac{\pi}{-2ik\sqrt{(X-x')^2 + Z^2}}}\exp\{ik\sqrt{(X-x')^2 + Z^2}\}\frac{Z}{\sqrt{(X-x')^2 + Z^2}}$$

For $Z \gg X$, we can treat $\sqrt{(X-x')^2 + Z^2}$ as Z in the denominator. However, for the phase factor, we need to expand it in powers of x' and obtain

$$\sqrt{(X-x')^2 + Z^2} = Z + \frac{(X-x')^2}{2Z} + \cdots$$

Technically, we cannot drop higher order terms, as the integration in x' extends to infinity. But in reality, there is no monochromatic wave with infinite spatial extent. Following this argument, we can assume that there is a decay factor in y', which justifies the expansion. By the same argument, we can also treat the y' integration as a gaussian one and reach the same result as before. Then,

$$\psi(X,0,z,t) = \frac{k}{\pi i}\int_0^{+\infty} dx' \sqrt{\frac{\pi}{2}}\,\frac{1+i}{\sqrt{2}}\sqrt{\frac{1}{kz}}\, \exp\{ik[z+(X-x')^2/2z]\}\, e^{-i\omega t}\, I_0^{1/2}$$

Let $x' = u + (k/2z)^{1/2}X$, we can express the integral as

$$\psi(X,0,z,t) = I_0^{1/2} e^{ikz-i\omega t}\,\frac{k}{\pi i}\sqrt{\frac{\pi}{2}}\,\frac{1+i}{\sqrt{2}}\,\frac{1}{\sqrt{kz}}\sqrt{\frac{2z}{k}}\int_{-\xi}^{+\infty} e^{iu^2}\,du$$

$$= I_0^{1/2} e^{ikz-i\omega t}\left(\frac{1+i}{2i}\right)\sqrt{\frac{2}{\pi}}\int_{-\xi}^{+\infty} e^{iu^2}\,du,$$

with $\xi = (k/2z)^{1/2}X$.

1b) The intensity is given by

$$I = |\psi|^2 = \frac{I_0}{2}\left|\sqrt{\frac{2}{\pi}}\int_{-\xi}^{+\infty}[\cos(u^2) + i\sin(u^2)]\,du\right|^2$$

By the definition of the cosine integral,

$$C(z) = \sqrt{\frac{2}{\pi}}\int_0^z \cos(u^2)\,du,$$

we have

$$\sqrt{\frac{2}{\pi}}\int_{-\xi}^{+\infty}\cos(u^2)\,du = \sqrt{\frac{2}{\pi}}\int_0^{+\infty}\cos(u^2)\,du + \sqrt{\frac{2}{\pi}}\int_{-\xi}^{0}\cos(u^2)\,du$$

$$= \sqrt{\frac{2}{\pi}}\int_0^{\xi}\cos(u^2)\,du + \frac{1}{2} = C(\xi) + \frac{1}{2},$$

where we have used the result that

$$\int_0^{+\infty}\cos(u^2)\,du = \int_0^{+\infty}\sin(u^2)\,du = \frac{\sqrt{2\pi}}{4}.$$

Therefore, $I = \frac{I_0}{2}\left[\left(C(\xi)+\frac{1}{2}\right)^2 + \left(S(\xi)+\frac{1}{2}\right)^2\right]$

(c) From Eq (10.101), the diffracted field is

$$\vec{E}(\vec{x}) = \frac{1}{2\pi}\nabla\times\int(\vec{n}\times\vec{E})\frac{e^{ikR}}{R}\,da'.$$

Here, for the incident electric field, its polarization is in the xy plane,

$$\vec{E}_i(\vec{x}) = E_0(\vec{\varepsilon}_1\cos\beta + \vec{\varepsilon}_2\sin\beta)\,e^{ikz}$$

Then $(\vec{n}\times\vec{E}_i)_{z=0} = E_0(\vec{\varepsilon}_2\cos\beta - \vec{\varepsilon}_1\sin\beta)$. Also, $\vec{R} = (X-x', -y', z)$. Ignore slowly varying terms, we have

$$\vec{E}(\vec{x}) = \frac{ik}{2\pi}\int\frac{\vec{R}\times(\vec{n}\times\vec{E})}{R^2}e^{ikR}\,da'$$

$$= \frac{ikE_0}{2\pi}\int\frac{1}{R^2}\left[\vec{\varepsilon}_1(-Z\cos\beta) + \vec{\varepsilon}_2(-Z\sin\beta) + \vec{\varepsilon}_3\left((X-x')\cos\beta - y'\sin\beta\right)\right]$$

$$\times\exp\left\{ik\left(Z+\frac{y'^2}{2Z}+\frac{(X-x')^2}{2Z}\right)\right\}dx'dy'$$

For $Z \gg X$, we can write the result as

$$\vec{E}(\vec{x}) = -\frac{ikE_0}{2\pi}\frac{e^{ikZ}}{Z}\sqrt{\frac{2i\pi Z}{k}}\left(\vec{\varepsilon}_1\cos\beta + \vec{\varepsilon}_2\sin\beta\right)\int_0^{+\infty}\exp\left\{ik\frac{(X-x')^2}{2Z}\right\}dx'$$

$$= \vec{E}_i\,e^{ikz}\left(\frac{1+i}{2i}\right)\sqrt{\frac{2}{\pi}}\int_{-\xi}^{+\infty}e^{it^2}dt,$$

Except for the vector nature of the expression, it is identical to the scalar result.